CDT204 – Computer Architecture

The Exam has 65 points the grades will be awarded as follows:

- 3: 33 points
- 4: 46 points
- 5: 59 points

Important Notes:

- Give as full an answer as possible to obtain full marks. All calculations, approximations, assumptions and justifications must be reported for full credit unless stated otherwise. Please use figures and examples to clarify.
- If you do not understand a question clearly, you are allowed to call the teacher and ask.
- Write the question and part number on each page clearly.
- Answer each question on a separate page.

1. (6 points) Flynn's taxonomy in its classical form (represented in the table below) defines 4 different classes in computer architecture from a multiprocessing point of view.

	Single Instruction	**Multiple Instruction**
Single Data	SISD	MISD
Multiple Data	SIMD	MIMD

   (a) (4 points) Explain each of the classes and give examples for each of them.

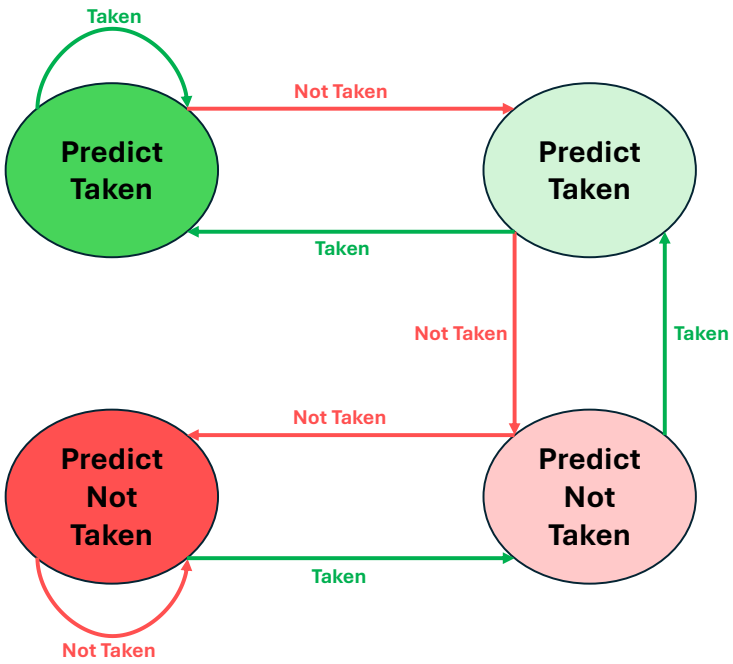
   (b) (2 points) Some researchers have proposed 2 newer classes called SPMD (Single Program Multiple Data) and MPMD (Multiple Programs Multiple Data). SPMD defines a design in which multiple processors execute the same program at independent points on different data at the same time. Can you give an example of hardware which has an architecture very similar to SPMD?

2. (9 points) Some values from the table below are removed and marked with red placeholder characters (a-s). For each placeholder present the correct value that should be in that place. Assume 32-bit byte-addressed memory addresses.

	Total Size	**Block Size**	**No. Ways**	**No. Sets**	**No. Bits (Tag/Index/Offset)**	**Cache Organization**
Cache A	2^{14} Bytes	**a**	2	256	(**b** / **c** / **d**)	**e**
Cache B	32 KB	64	**f**	64	(**g** / **h** / **i**)	**j**
Cache C	**k** KB	64	64	1	(**l** / **m** / **n**)	**o**
Cache D	8 KB	**p**	**q**	**s**	(19/ 6/ 7)	Direct mapped

   Note: By "Cache Organization" we mean the term which is normally associated with the organization of information in the cache.

3. (8 points) Assume the following repeating pattern (for example in a loop) of branch outcomes: T, NT, T, T, NT (T: Taken, NT: Not Taken)



(a) (2 points) What is the accuracy of always-taken and always-not-taken predictors for this sequence of branch outcomes?

(b) (2 points) What is the accuracy of the two-bit predictor for the first 4 branches in this pattern, assuming that the predictor starts off in the bottom left state (predict not taken) of the above diagram?

(c) (1 point) What is the accuracy of the two-bit predictor if this pattern is repeated forever?

(d) (3 points) Design a predictor that would achieve a perfect accuracy if this pattern is repeated forever.

4. (7 points) For two direct-mapped cache designs with a 32-bit address, the following bits of the address are used to access the cache.

Cache	**Tag**	**Index**	**Offset**
A	31-10	9-5	4-0
B	31-12	11-6	5-0

Starting from power on, the following byte-addressed cache references are recorded:

Address	**0**	**4**	**16**	**132**	**232**	**160**	**1024**	**430**	**140**	**3100**	**180**	**2180**

(a) (1 point) What is the cache line size (in words) for caches A and B?

(b) (1 point) How many entries does each cache have?

(c) (2 points) For each cache, what is the ratio of total implementation bits to data storage bits? Assume one valid bit per cache line.?

(d) (2 points) How many blocks are replaced if Cache A is used?

(e) (1 point) What is the hit ratio for Cache A?

5. (6 points) Consider the following MIPS program

```
.globl __start

.data
x: .word 256
.text
__start:
        addi $s0, $zero, 0
        addi $s1, $zero, 1
        lw   $s2, x
loop:
        beq  $s2, $s1, end
        div  $s2, $s2, 2
        addi $s0, $s0, 1
        j    loop
end:
        ...
```

(a) (3 points) What will the value of $s0 register be when the program reaches point "end"? (The 'DIV' instruction performs integer division). Motivate your answer.

(b) (3 points) Write the equivalent C code for the above MIPS code.

6. (10 points) The table below shows the number of instructions per processor core on a multicore processor as well as the average CPI for executing the program on 1, 2, 4, or 8 cores.

Processors	**No. Instructions per Processor**	**CPI**
1	1.00E+10	1.2
2	5.00E+09	1.4
4	2.50E+09	1.8
8	1.25E+09	2.6

Assume that the power consumption of a processor core can be described by the following equation:

$$Power = CfV^2$$

where f is the frequency and C is a constant which in this case it is equal to 5 mA/MHz (the core needs 5 milliamps per MHz of frequency). V is the operation voltage of the processor and can be described by the following equation:

$$Voltage = \frac{1}{5} frequency + 0.4$$

with the $frequency$ measured in GHz. So, for example, at 5 GHz, the voltage would be 1.4 V.

(a) (2 points) Assuming a 3 GHz clock frequency, what is the execution time of the program using 1, 2, 4, or 8 cores?

(b) (3 points) What should the CPI of a single core CPU be to get an execution time equal to the time obtained by each of the multicore options in the table above. Note that the number of instructions should be the aggregate number of instructions executed across all the cores.

(c) (3 points) Find the power consumption of the program executing on 1, 2, 4, and 8 cores assuming that each core is operating at a 3 GHz clock frequency. Likewise, find the power consumption of the program executing on 1, 2, 4, or 8 cores assuming that each core is operating at 500 MHz.

(d) (2 points) Discuss the results from part C. What can you conclude from the data?

7. (7 points) For a direct-mapped cache design with a 32-bit address, the following bits of the address are used to access the cache

Tag	**Index**	**Offset**
31-12 (20 bits)	11-5 (7 bits)	4-0 (5 bits)

Assume the following C program is run

```
int arr[2048];
void main(){
        arr[4]    = 20;         // 1
        arr[1029] = 8;          // 2
        arr[7]    = 2008;       // 3
        arr[9]    = arr[4];     // 4
        arr[10]   = arr[9] + 7; // 5
}
```

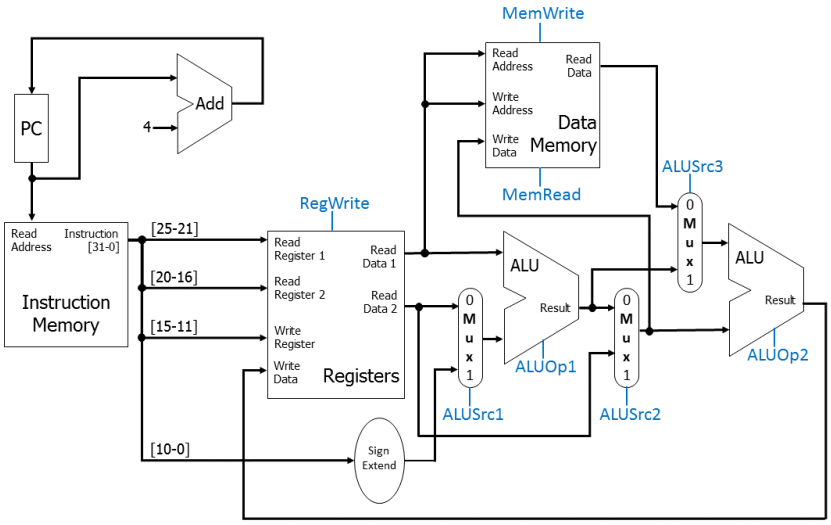
Suppose that the array is allocated by the compiler in a way that it starts at an address divisible by 4096 and that the array is the only thing that will occupy the space in the data cache. Also assume that an `int` takes up 4 bytes of space.

(a) (2 points) What is the cache block size (in words)?

(b) (1 point) How many entries does the cache have?

(c) (1 point) What is the cache size?

(d) (3 points) Assuming that the cache is direct mapped and starts empty, which of the memory accesses from the program will generate a cache miss and why?



8. (12 points) The datapath above supports the following instructions:

   - `lw_add rd, (rs), rt` (fused load and add): Register `rd` will be set to the summation of memory value addressed by register `rs` and the value stored in register `rt`. `R[rd] = Mem[R[rs]]+R[rt]`
   - `addi_st (rs), rs, imm` (fused immediate add and store): The memory addressed by register `rs` will be set to summation of register `rs` and constant value `imm`. `Mem[R[rs]]=R[rs]+imm`
   - `sll_add rd, rs, rt, imm` (fused immediate left shift and sum): Value of register `rs` is shifted left by constant value `imm` then added to register `rt` and stored in register `rd`. `R[rd]=(R[rs]«imm)+R[rt]`

   All instructions use the same format (shown below). Not all instructions use all the fields below.

op	**rs**	**rt**	**rd**	**imm**
(bits 31-26)	(bits 25-21)	(bits 20-16)	(bits 15-11)	(bits 10-0)

   For each of the above instructions, specify how the control signals should be set for correct operation. `ALUOp` can be `ADD`, `SUB`, `SLL`, `PASS_A`, or `PASS_B` (i.e. `PASS_A` means pass the top operand through without changing it). Use `X` for don't care.

   You can find an empty table for this task at the last page of the exam sheet, which you can use to write your answer in. Detach it from the exam sheet and put it in your answers folder.

Inst.	ALUSrc1	ALUSrc2	ALUSrc3	ALUOp1	ALUOp2	MemRead	MemWrite	RegWrite
lw_add								
addi_st								
sll_add								